

1 INTRODUCTION

1.1 PROJECT DESCRIPTION

Due to people's heavy reliance on bikes for daily transportation two-wheeler accidents have grown to be a serious concern. Many of these incidents worsen because the rider was not wearing a helmet or did not receive assistance in a timely manner rather than because of the crash itself. Even though these issues are straightforward they have significant repercussions. The purpose of this project is to address these common problems that we encounter on a daily basis.

The Smart Helmet System was designed with the intention of making riding safer without making it more difficult for the rider. This is a very straightforward concept. For the purpose of determining whether or not the rider is actually wearing the helmet a small sensor is installed inside the helmet. Should the helmet not be worn in the correct manner the bicycle will not start. In order to guarantee that the rider adheres to the fundamental safety rule before even beginning the journey this straightforward step is taken.

The identification of accidents is yet another essential component of this project. The MPU6050 is an example of a motion sensor that performs continuous checks to identify any unusual movements sudden falls or significant impacts. In the event that such an occurrence takes place the system will presume that an accident has taken place. Additionally at that precise moment the GPS module will send an emergency location to the contact that the rider has saved. As a result this is helpful in circumstances in which the rider may be unconscious or unable to call for assistance. Because even a brief delay can have a significant impact the automatic alert has the potential to save a significant amount of time.

In addition the project incorporates a safety feature that will cause the bicycle to come to a stop in the event that the rider loses their balance while riding. Through the use of microcontrollers such as ESP32 communication is established between the helmet unit and the bike unit in order to accomplish this. Because of this the bike is prevented from continuing to move without being controlled which would otherwise result in additional damage.

The overarching objective of this Smart Helmet project is to create something that is not only straightforward but also inexpensive and genuinely helpful. The focus is on real-world issues that riders encounter and it makes an effort to find solutions to those issues by utilizing readily available electronic components. This system can help reduce the number of injuries that are caused by accidents and encourage people to ride them in a more responsible manner by combining fundamental Internet of Things technology with practical safety needs.

❖ Problem Statement

Despite advancements in road safety awareness two-wheeler riders still have a high risk of suffering severe injuries in collisions because many riders still don't wear helmets correctly or don't get help quickly after a fall. Despite being the most straightforward and effective safety precaution helmets are frequently disregarded by riders and there is currently no system in place to guarantee that a helmet is worn before starting the vehicle which encourages risky riding practices and raises the possibility of serious head injuries. It can be challenging to get emergency assistance in many accident situations particularly in isolated or sparsely populated areas. This is because people in the vicinity might not notice the incident or might not be able to react promptly which causes delays in traditional accident reporting that relies solely on bystanders. The rider's condition may deteriorate as a result of these delays and in certain situations the bike may continue to move after the rider has fallen endangering both the rider and other drivers. As a result there is a great need for an easy-to-use reasonably priced and automated safety system that can detect whether the helmet is worn correctly recognize unexpected falls or collisions in real time send emergency alerts without the need for human intervention and stop the vehicle from moving without the rider. The proposed Smart Helmet System will make two-wheeler travel much safer and more dependable by enforcing helmet use enhancing accident detection and ensuring prompt emergency communication.

1.2 Objectives of the Study

- The development of a Smart Helmet system that is both straightforward and easy to use with the capability of automatically determining whether the rider is wearing the helmet correctly prior to starting the vehicle. This will ensure that fundamental safety regulations are adhered to without the need for manual inspection.
- In order for the system to accurately identify helmet status detect sudden falls or accidents and send emergency notifications to predefined contacts it is necessary to integrate essential Internet of Things-based sensors and modules. These include infrared sensors MPU6050 GSM/SMS alert systems and microcontrollers such as ESP32.

- To ensure the dependability and efficiency of the accident detection mechanism by testing the system under a variety of conditions with the objective of achieving consistent performance a quick response time and a minimum of false triggers while the movement is occurring in real time.
- To provide immediate and meaningful safety actions during accidents including sending an alert message to emergency contacts with the necessary details and stopping the vehicle when the rider falls thereby improving the chances of timely assistance and reducing further risk.

1.3 Scope of the Project

The goal of this project is to create a useful and user-friendly Smart Helmet system that prioritizes rider safety by guaranteeing helmet-wearing detection accident identification and automated emergency communication. In addition to monitoring the rider's movement in real time using sensors to identify falls or abrupt impacts the system attempts to prevent the vehicle from starting if the helmet is not worn correctly. The system instantly notifies the rider's emergency contacts in the event of an accident enabling them to get help in a timely manner.

Future additions to the project could include voice alerts GPS-based accident location tracking mobile app connectivity enhanced wireless communication between helmets and bike and integration with traffic safety systems. Accuracy and dependability can also be improved by adding more sophisticated sensors and modules. All things considered the project provides the fundamental safety features needed for two-wheeler riders and it may be improved further to produce a more sophisticated and all-encompassing road-safety solution.

2. LITERATURE SURVEY

2.1 EXISTING AND PROPOSED SYSTEM

The majority of two-wheeler riders currently rely solely on manual safety procedures and simple helmets which have a number of drawbacks. Although helmet use is encouraged by traffic laws and road safety campaigns there is no automated system that guarantees riders put on the helmet before starting their bike. Conventional helmets lack any technology to track rider safety identify collisions or communicate emergencies. In the event of an accident assistance is frequently provided by bystanders which causes delays in getting medical attention particularly in isolated locations where prompt assistance is less likely.

The majority of accident detection systems on the market today are costly and typically made for cars rather than two-wheelers. Some simple solutions use mobile applications to identify falls but these rely largely on the phone's sensors and necessitate the user to maintain the mobile device in a fixed position which isn't always feasible. Additionally these systems are unreliable for use in the real world because they are unable to determine whether the rider is wearing the helmet correctly.

Additionally some early Smart Helmet system prototypes developed in research projects employ simple sensors like switches or pressure sensors to detect helmet wear. Nevertheless these systems are often limited to a single purpose like preventing vehicle ignition without the helmet or detecting falls without any communication features. They don't provide a single system that combines helmet detection emergency messaging vehicle control and accident detection. Another disadvantage of existing solutions is their lack of useful and intuitive designs. Because many prototypes require heavy parts or complex wiring they are difficult to use in real-world scenarios. Because they depend on short-range wireless modules that might not function well over longer distances some systems are less dependable. Furthermore most accident-detection devices in use today do not automatically notify emergency contacts so the rider must manually report the incident—which may not be possible during serious accidents.

While a number of studies and prototypes have investigated various safety mechanisms such as the use of IR sensors for helmet status GPS modules for alerting or accelerometers for fall detection they frequently concentrate on just one problem at a time. There is no integrated

system that guarantees the use of helmets accurately detects collisions stops the bike when necessary and offers immediate communication assistance. These gaps demonstrate the need for a comprehensive Smart Helmet solution that addresses the main safety issues that two-wheeler users currently face while being easy to use affordable and dependable for regular riders.

➤ **Limitations of the System**

- **Limited Safety Features:** The current system primarily concentrates on vehicle ignition control emergency SMS alerts accident identification and helmet-wearing detection. This version lacks other sophisticated features like voice alerts camera monitoring GPS-based location tracking and traffic system integration.
- **Reliance on Sensor Precision:** Sensors like the MPU6050 and IR sensor are crucial to the Smart Helmet. The system may send out false alarms or fail to correctly identify an accident if these sensors are misaligned vibrate from the outside world or have calibration problems. Sensor performance may also be impacted by environmental elements like pressure or dust.
- **Short-Range Communication:** Wireless modules with a restricted range are used for communication between the bike unit and the helmet unit. The signal may not reach the bike correctly if there is interference or if the distance is greater than the supported limit which could have an impact on ignition control or system dependability.
- **Absence of Large-Scale Field Testing:** The Smart Helmet has not yet been put to the test in a variety of real-world riding scenarios including long-distance rides high-speed travel rough roads and heavy rain. It's unclear how reliably the system will function in various settings without thorough field validation.

2.2 PROPOSED SYSTEM

By integrating helmet-wearing detection accident identification emergency alert messages and vehicle ignition control into a single cohesive platform the suggested system presents an Internet of Things-based Smart Helmet intended to increase the safety of two-wheeler riders. This system integrates several safety modules to provide a more useful and dependable solution for real-world use in contrast to the current setups that primarily concentrate on a single feature such as a basic switch for helmet detection or a basic fall sensor.

A sensor inside the suggested Smart Helmet determines whether the rider is wearing it correctly. The system ensures the enforcement of fundamental safety regulations without requiring manual verification by only permitting the bike to start when the helmet is correctly worn. In addition the system tracks the rider's movement using an accelerometer and gyroscope (MPU6050). The system treats any abrupt fall forceful impact or unusual tilt as an accident and sends out an alert right away.

The Smart Helmet automatically sends an emergency SMS to the rider's registered contact number when it detects an accident by communicating with a GPS. In the event that no one is present at the scene of the accident this enables the rider to receive prompt assistance. The microcontroller modules like the ESP32 or control the communication between the bike unit and the helmet and make sure that the vehicle stops if the rider falls off. This stops the bike from moving further and lowers the possibility of further damage.

The entire system is inexpensive lightweight and simple to use. The system performs all functions including helmet verification accident detection and alert messaging without requiring any technical expertise from the rider. More sophisticated features like GPS location tracking mobile app connectivity real-time monitoring and cloud integration could be added to the Smart Helmet in the future. The system would be even more helpful with these additions particularly in high-risk or remote locations where quick assistance is not always available.

The overall goal of the suggested system is to provide two-wheeler riders with an intelligent dependable and user-friendly safety solution. This Smart Helmet helps promote safer riding practices facilitates faster emergency response and tackles the main safety issues encountered in typical road conditions by integrating several IoT components and crucial safety mechanisms into a single device.

➤ **Advantages:**

- **Multiple Safety Features in One System:** Unlike current configurations that concentrate on a single safety function the Smart Helmet integrates helmet-wearing detection accident identification emergency SMS alerts and vehicle ignition control on a single platform.
- **Automatic and User-Friendly Operation:** Without the rider's technical expertise the system operates automatically. Anyone can easily use it because emergency messaging accident sensing and helmet detection all operate in the background.
- **Real-Time Accident Response:** When a fall or crash is detected the system instantly notifies the rider's emergency contacts facilitating a prompt response and minimizing delays in getting medical help.
- **Improved Rider Security:** The system encourages responsible riding and helps lower the risk of head injuries during collisions by making sure the bike cannot start without the helmet on.
- **Cost-Effective Solution:** The Smart Helmet is inexpensive and appropriate for widespread use without costly installations because it is constructed using basic IoT components like IR sensors MPU6050 ESP32 and GPS modules.
- **Scalable and Expandable Architecture:** More features such as GSM location tracking mobile app integration theft protection or voice alerts can be added in the future without major redesign thanks to the modular structure of the system.
- **Operates Without the Internet:** The system is dependable even in remote or low-network areas because the SMS alert is sent via a GSM module which eliminates the need for an internet connection.
- **Low Maintenance Design:** Since each module—such as bike control fall detection and helmet detection—is managed independently updates repairs and replacements are simpler and easier to handle.
- **Raises Awareness of Road Safety:** The system encourages riders to regularly adhere to safety regulations by automatically enforcing helmet usage thereby indirectly raising awareness of the significance of protective gear.
- **Avoid Secondary Mishaps:** When a rider falls the bike stops or the ignition cuts off lowering the possibility of the vehicle moving erratically and averting further damage or injuries.

2.3 FEASIBILITY STUDY

Before beginning full-scale development a feasibility study is conducted to ascertain whether the suggested IoT-based Smart Helmet system is workable implementable and capable of providing actual benefits. In order to make sure that the system can be successfully constructed tested and used in real-world situations this study looks at a number of factors including technical feasibility economic feasibility and operational feasibility. The objective is to verify that the features of the Smart Helmet including helmet-wearing detection accident identification emergency SMS alerts and vehicle ignition control can be reliably integrated using current technologies and that the project is worthwhile from a usability and cost standpoint.

❖ Technical Feasibility:

Because the project was created using dependable readily available and well-supported microcontroller and Internet of Things components it is technically feasible. The system makes use of widely used inexpensive and well-documented hardware including the ESP32 MPU6050 accelerometer and gyroscope IR sensor and GPS. These elements offer a reliable setting for the implementation of wireless communication emergency alert features accident identification and helmet-wearing detection.

The ESP32 microcontroller platform is appropriate for creating real-time safety systems because it facilitates sensor interfacing wireless communication and data processing. The system can reliably identify falls or crashes thanks to the MPU6050 sensor's ability to measure tilt acceleration and impact levels. The GPS module improves usefulness in remote locations by enabling SMS alerts without requiring internet connectivity.

High-end hardware is not necessary for the system to function; lightweight modules basic wiring and simple power sources are sufficient. Because of this the solution can be easily integrated into a standard helmet and deployed on a small scale. Development debugging and upgrades are made easier by the components' strong compatibility with the Arduino IDE and other open-source programming environments. Expensive proprietary tools and complicated machinery are not required because all sensors and modules are open-source or widely supported. The design is modular meaning individual components can be replaced or modified without redesigning the entire system. All things considered the Smart Helmet

system can be developed implemented and maintained successfully thanks to the availability of IoT components straightforward programming tools and cross-platform compatibility.

❖ **Economic Viability:**

All of the sensors and modules used in the design are readily available in the market at reasonable prices which lowers overall expenses and the Smart Helmet system is economically viable because it was created using inexpensive IoT components and open-source platforms that don't require licensing fees. Without expensive equipment or specialized hardware development can be done with common microcontrollers and simple electronic tools. This project is affordable and can be implemented with little investment when compared to commercial smart safety solutions or advanced vehicle monitoring systems making it appropriate for both academic projects and useful real-world applications.

❖ **Operational Feasibility:**

The project is operationally feasible because it offers a straightforward and automated system that makes it simple for non-technical users like everyday riders or commuters to operate the helmet without any special knowledge. Once powered on it performs functions like helmet detection accident sensing and emergency messaging automatically ensuring that it can be used effectively by the intended users and supporting real-world safety needs by providing prompt responses during accidents.

The system can be installed on any standard helmet or two-wheeler without the need for special infrastructure because it is lightweight and device-independent guaranteeing dependable and seamless operation during daily travel.

Potential challenges include sensor sensitivity problems riders' lack of knowledge about smart safety devices or user reluctance brought on by mistrust of automated safety systems. It can be integrated into basic vehicle systems for better rider protection or used for personal safety. It is adaptable for both individuals and transportation services.

2.4 TOOLS AND TECHNOLOGIES USED

The technologies and tools utilized in this project were chosen to guarantee dependability speed and simplicity of use. While sensors like the IR sensor detect helmet wear and the MPU6050 handles accident and fall detection microcontrollers like the ESP32 handle sensor inputs and wireless communication. The system works even in the absence of internet access because the GPS is used to send emergency SMS alerts. Hardware integration is supported by rechargeable power sources jumper wires and basic electronic components. The Smart Helmet system is practical for everyday use economical and efficient thanks to these technologies.

V S Code

A popular lightweight and effective source code editor for web and Internet of Things development Visual Studio Code (VS Code) was created by Microsoft. Using HTML CSS and JavaScript the admin-based web application in this project was developed and managed using Visual Studio Code. It enables the administrator to view the current locations of all riders and send emergency messages when necessary. With features like syntax highlighting auto-completion error detection and debugging support Visual Studio Code offered a tidy and well-organized environment that reduced coding errors and increased productivity. Its built-in terminal and Git support made project management simpler and its Live Server extension was helpful for testing frontend changes in real time. All things considered Visual Studio Code provided a quick adaptable and developer-friendly environment for creating and All things considered VS Code provided a quick adaptable and developer-friendly environment for creating and managing the Smart Helmet system's web interface.

Studio for Android

The Smart Helmet system's Android mobile app was made with Android Studio. The app is in charge of showing alerts rider status and emergency messages that come from the helmet and cloud services. Android Studio provided tools for UI design Java/Kotlin coding debugging and testing on real devices. Its emulator and logcat features helped identify and fix application-level issues efficiently. Android Studio played a key role in creating a reliable and user-friendly mobile interface for monitoring helmet safety events.

Blynk Application

The Blynk IoT platform was used to build a real-time monitoring and control dashboard for the Smart Helmet. Through the Blynk mobile app sensor data such as accident detection status and GPS location is visualized in real time. Blynk also enables instant notifications and alerts when a fall or emergency is detected. The platform simplified IoT communication between the ESP32 and the mobile interface without complex backend development making it ideal for rapid prototyping and real-time monitoring.

Firebase

Firebase was used as the cloud backend for storing and syncing real-time data such as rider locations emergency alerts and system status. Its real-time database ensures that updates from the helmet and mobile app are instantly reflected on the admin web dashboard. Firebase provides secure data storage fast synchronization and easy integration with both web and Android applications. This makes it suitable for handling live tracking and emergency communication in the Smart Helmet project.

HTML

HTML (Hypertext Markup Language) was used to structure the admin web dashboard including pages for viewing rider locations alert messages and system status. It provides the basic layout and content structure required for the web interface.

CSS

CSS (Cascading Style Sheets) was used to style the web application and make the admin dashboard visually clear and user-friendly. It helped in designing responsive layouts proper color schemes and readable interfaces for monitoring multiple riders.

JavaScript

JavaScript was used to add interactivity and real-time functionality to the web application. It enables fetching data from Firebase updating rider locations dynamically displaying alerts and sending emergency messages from the admin dashboard. JavaScript ensures smooth interaction without reloading the page and improves the overall responsiveness of the system.

2.5 HARDWARE AND SOFTWARE REQUIREMENTS

A. Hardware Requirements

- ESP32 Microcontroller
- Accelerometer (MPU6050)
- IR Sensor
- Limit switches
- Cell
- GPS Module with Antenna
- Driver Circuit Board
- DC Motor
- Mobile Phone
- Helmets
- Jumper Wires

B. Software Requirements

Requirements	Specification
Operating System	Windows 10 or above
Programming Languages	HTML CSS JavaScript Embedded C (for ESP32)
Development Tools / IDE	Visual Studio Code (for web development) Android Studio (for Android application)
IoT Platform	Blynk Application (for real-time monitoring)
Cloud Backend	Firebase (for real-time database and data synchronization)
Web Technologies	HTML5 CSS3 JavaScript (for admin dashboard)
Mobile Platform	Android OS (for alert and monitoring application)

3. SOFTWARE REQUIREMENTS SPECIFICATION

3.1 USERS REQUIREMENTS

Riders (Helmet Users)

Riders are the primary users of the Smart Helmet system. They wear the helmet while riding and the system continuously monitors their safety using sensors. In case of an accident or fall the helmet automatically detects the event and sends the rider's live location to the cloud. Riders benefit from automatic emergency alerts without the need for manual action ensuring faster help during critical situations. The system improves rider safety by providing real-time monitoring and quick emergency response.

Administrator (Monitoring Authority)

The administrator is responsible for monitoring all riders through the web dashboard and mobile application. The admin can view the live location of all registered riders receive accident alerts and send emergency messages when required. The administrator uses the system to track rider status in real time and coordinate timely assistance during emergencies. Secure access ensures that only authorized personnel can monitor data and communicate with riders.

Emergency Contacts / Family Members

Emergency contacts or family members receive alerts when an accident is detected. They are notified with the rider's location and emergency information through the mobile application or messaging system. This allows family members to respond quickly and provide assistance or contact emergency services when needed. Their role is limited to receiving alerts and supporting the rider during emergencies.

System Developers / Maintainers

System developers and maintainers are responsible for designing developing and maintaining the Smart Helmet system. They manage the ESP32 firmware sensor integration Firebase cloud services web dashboard Android application and Blynk platform. Developers ensure smooth data flow between the helmet cloud and user

interfaces. They also fix bugs improve system performance and ensure data security and reliability. Their work guarantees that the system runs continuously and adapts to future improvements.

3.2 FUNCTIONAL REQUIREMENTS

Accident / Fall Detection

- The system must continuously monitor rider movement using sensors mounted on the smart helmet.
- When a sudden impact or fall is detected the system must automatically identify it as a possible accident without manual input from the rider.

Live Location Tracking

- The system must obtain the rider's real-time location using the GPS module connected to the helmet.
- The live location data must be sent to the cloud and updated continuously.
- The administrator must be able to view the live location of all riders through the web dashboard or mobile application.

Emergency Alert Generation

- When an accident is detected the system must automatically generate an emergency alert.
- The alert must include the rider's current location and accident status.
- The alert must be sent to the administrator and emergency contacts without delay.

Admin Monitoring and Control

- The administrator must have secure access to the monitoring dashboard.
- The admin must be able to view rider status accident alerts and live location information.
- The admin must be able to send emergency messages or instructions to riders when required.

Mobile Application Alerts

- The system must display accident alerts and notifications on the Android mobile application.
- Notifications must be shown in real time whenever an emergency is detected.

Cloud Data Handling

- The system must store and update rider location alert status and helmet data in Firebase.
- All updates must be reflected instantly on the web dashboard and mobile app.

System Reliability

- The system must continue monitoring as long as the helmet device is powered on.
- The system must avoid false alerts as much as possible by validating sensor data before triggering emergency messages.

3.3 NON-FUNCTIONAL REQUIREMENTS

Security

- The system must ensure secure access to the admin dashboard and mobile application.
- Only authorized administrators should be able to view rider locations and send emergency messages.
- Cloud data stored in Firebase must be protected using proper authentication and access rules.

Usability

- The system interface must be simple clear and easy to understand for both riders and administrators.
- Alerts notifications and messages must be clearly displayed on the mobile application and web dashboard.
- The admin dashboard must allow easy navigation to monitor rider locations and emergency status.

Performance and Speed

- Accident detection must be processed immediately after a fall or impact is detected.
- Live location updates must be sent and displayed in near real time without noticeable delay.
- Notifications and alerts must be delivered quickly to ensure timely emergency response.

Reliability and Availability

- The system must operate reliably whenever the helmet device is powered on.
- Emergency alerts and location updates must work consistently without failure.
- The cloud platform must maintain continuous data synchronization between the helmet web dashboard and mobile app.

Scalability

- The system should support an increase in the number of riders without affecting performance.
- The cloud-based design should allow future expansion such as adding more helmets or additional monitoring features.

Maintainability

- The codebase should be well-structured and modular for easy understanding and updates.
- Developers should be able to fix bugs improve features and upgrade the system without affecting existing functionality.

3.4 HARDWARE SPECIFICATION

ESP32 Microcontroller with Inbuilt Wi-Fi Module

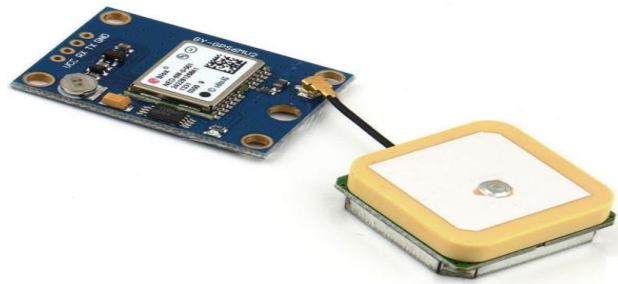
- A feature-rich MCU with integrated Wi-Fi and Bluetooth connectivity for a wide range of applications.

- ESP32 is capable of functioning reliably in industrial environments. Its operating temperature ranges from -40°C to $+125^{\circ}\text{C}$. Equipped with advanced calibration circuitries ESP32 can dynamically remove external circuit imperfections and adapt to changes in external conditions.
- Engineered for mobile devices wearable electronics and IoT applications ESP32 achieves ultra-low power consumption with a combination of several types of proprietary software. ESP32 also includes state-of-the-art features such as fine-grained clock gating various power modes and dynamic power scaling.
- ESP32 can perform as a complete standalone system or as a slave device to a host MCU. This reduces communication stack overhead on the main application processor. ESP32 can interface with other systems to provide Wi-Fi and Bluetooth functionality through its SPI / SDIO or I2C / UART interfaces.
- ESP32 modules can be ordered with different antenna configurations (e.g. PCB antenna antenna connector) and flash sizes so that they correspond to the needs of different applications. ESP32 modules also offer manufacturing customizations with the pre-programming of application firmware custom data and pre-provisioning with cloud certificates. With in-built antenna switches RF balun power amplifier low-noise receive amplifier filters and power management modules our chips add priceless functionality and versatility to your applications with minimal PCB requirements.



GPS Module

- Using the Global Positioning System (GPS all process used to establish all position at any point on the globe) the following two values can be determined anywhere on Earth. GPS receivers are used for positioning locating navigating surveying and determining the time and are employed both by private individuals (e.g. for leisure activities such as trekking balloon flights and cross-country skiing etc) and companies (surveying determining the time navigation vehicle monitoring etc) GPS (the full description is: Navigation System with Timing And Ranging Global Positioning System



Tilt sensor

- The MPU6050 is a small thin low power complete 3-axis accel-erometer with signal conditioned voltage outputs. The product measures acceleration with a minimum full-scale range of ± 3 g. It can measure the static acceleration of gravity in tilt-sensing applications as well as dynamic acceleration resulting from motion shock or vibration.
- The user selects the bandwidth of the accelerometer using the CX CY and CZ capacitors at the XOUT YOUT and ZOUT pins. Bandwidths can be selected to suit the application with a range of 0.5 Hz to 1600 Hz for the X and Y axes and a range of 0.5 Hz to 550 Hz for the Z axis.



Limit Switch

It is also used as bump sensor in accident detection projects and also used in Mechanical based projects where you have to determine if the object has reached a specific location. It is also used in CNC machines & 3D printer to detect if the part has reached the start or end point.

Features of Limit Switch

- Technical Name: Limit Switch/Bump Switch
- Dimension: 20 mm x 16 mm x 6.7 mm
- Operating Temperature: -25 to +65
- Operating Max Volt and Current: 5A 125V-3A 250V

Reasons to use a mechanical switch

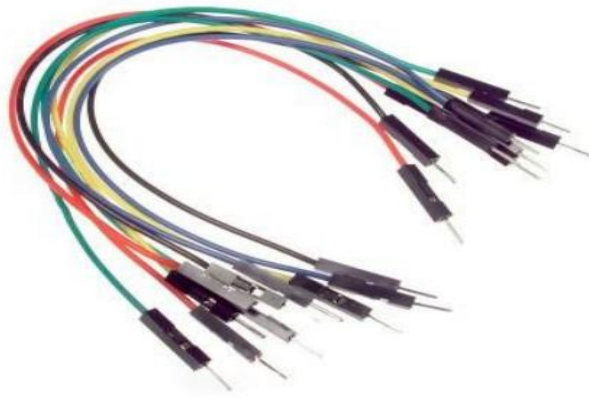
- Switches are the cheapest end stops in most cases.
- No need for opto pcb
- Simple switches can be used on x and y axis
- You could even make your own contact switch from a few pieces of metal.
- You get to solder stuff.



Jumper Wires

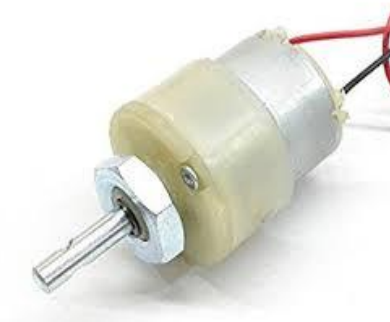
Jumper wires are used for making connections between items on your breadboard and your Adriano's header pins. Use them to wire up all your circuits. Jumper wires are simply wires that have connector pins at each end allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed fairly simple. In fact it doesn't get much more basic than jumper wires. Jumper wires typically come in three

versions: male-to-male ,male-to-female and female-to-female. The difference between each is in the end point of the wire. Male ends have a pin protruding and can plug into things while female ends do not and are used to plug things into. Male-to-male jumper wires are the most common and what you likely will use most often.



DC motors

- A DC motor is a mechanically commutated
- Electric motor powered from direct current (DC).
- The stator is stationary in space by definition
- The current in the rotor is switched by the commutator to also be stationary in space. This is how the relative angle between the stator and rotor magnetic flux is maintained near 90 degrees which generates the maximum torque.



4. SYSTEM DESIGN

4.1 System Perspective

This framework is helpful and the undesirable requirements are both meant to be met during this process of the actualization. And Includes key project elements such as the system design and the interfaces and users and data.

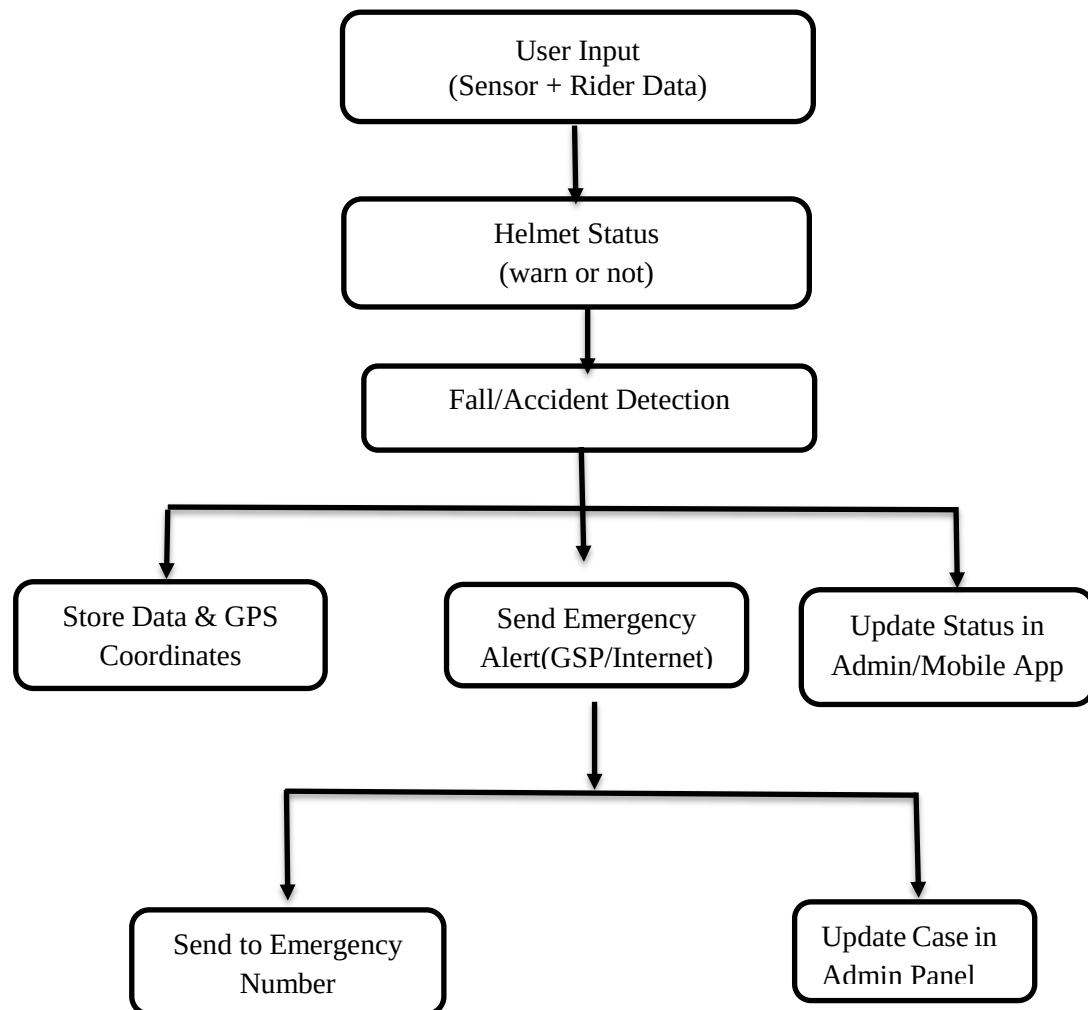


Fig 4.1: System Perspective of Smart Helmet

A system perspective is provided by combining multiple components to enable effective accident detection rider safety monitoring and emergency alert notifications for the timely assistance and protection of riders. This system perspective offers a comprehensive understanding of how the Smart Helmet project functions.

4.2 DATAFLOW DIAGRAM / CONTEXT DIAGRAM

A description of the system or the process being modeled can be found in a model diagram, commonly referred to as model level data flow diagram called (DFD). It shows how Sensors systems communicate with outsiders such as users and other systems or the organizations. The primary purpose of a texture diagram is to show the boundaries, dimensions, and relationships between the system and other objects under study. The survey diagram shows the changes in information such as it passes through the system. It graphically displays Sensor data with anything changes resulting from the context of the input.

- **Level: 0**

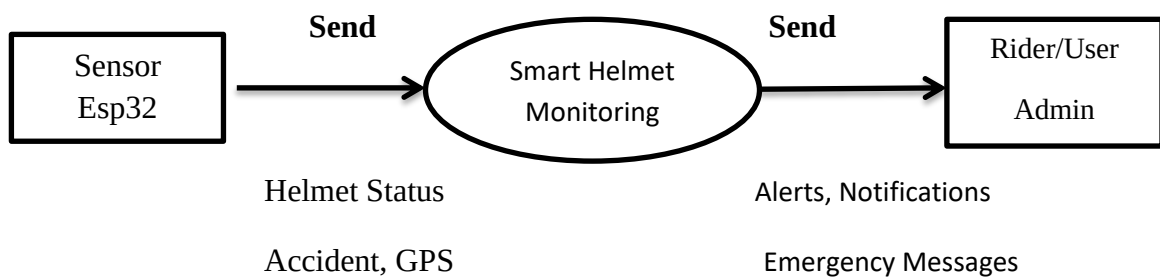


Fig 4.2: Context Diagram of Smart Helmet Monitoring System

This diagram gives a clear picture of how data moves through the various parts of the Smart Helmet System and how everything is structured. It emphasizes the important functions of key modules, like detecting when the helmet is being worn and identifying accidents. Additionally, the diagram illustrates how the system interacts with the mobile app and website to deliver real-time alerts, status updates, and accident notifications to users.

- **Level 1**

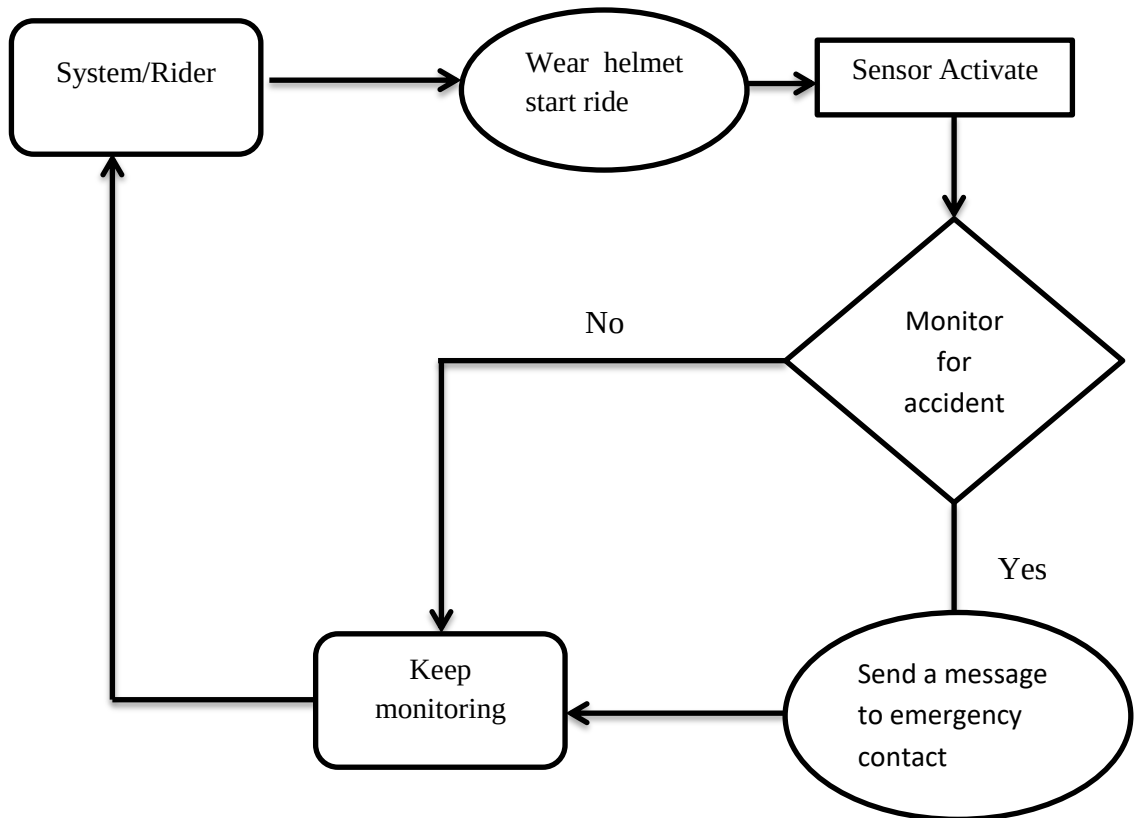


Figure 4.2.2 Level-1 DFD of Smart Helmet Monitoring System

The project's overall workflow showcases how the Smart Helmet System connects with the rider. It gathers sensor data about whether the helmet is being worn and detects any accidents. The system then processes this information and, based on its analysis, creates relevant status updates and accident alerts. These notifications are sent to the user via a mobile app and a web interface.

5. DETAILED DESIGN

5.1 Use case diagram

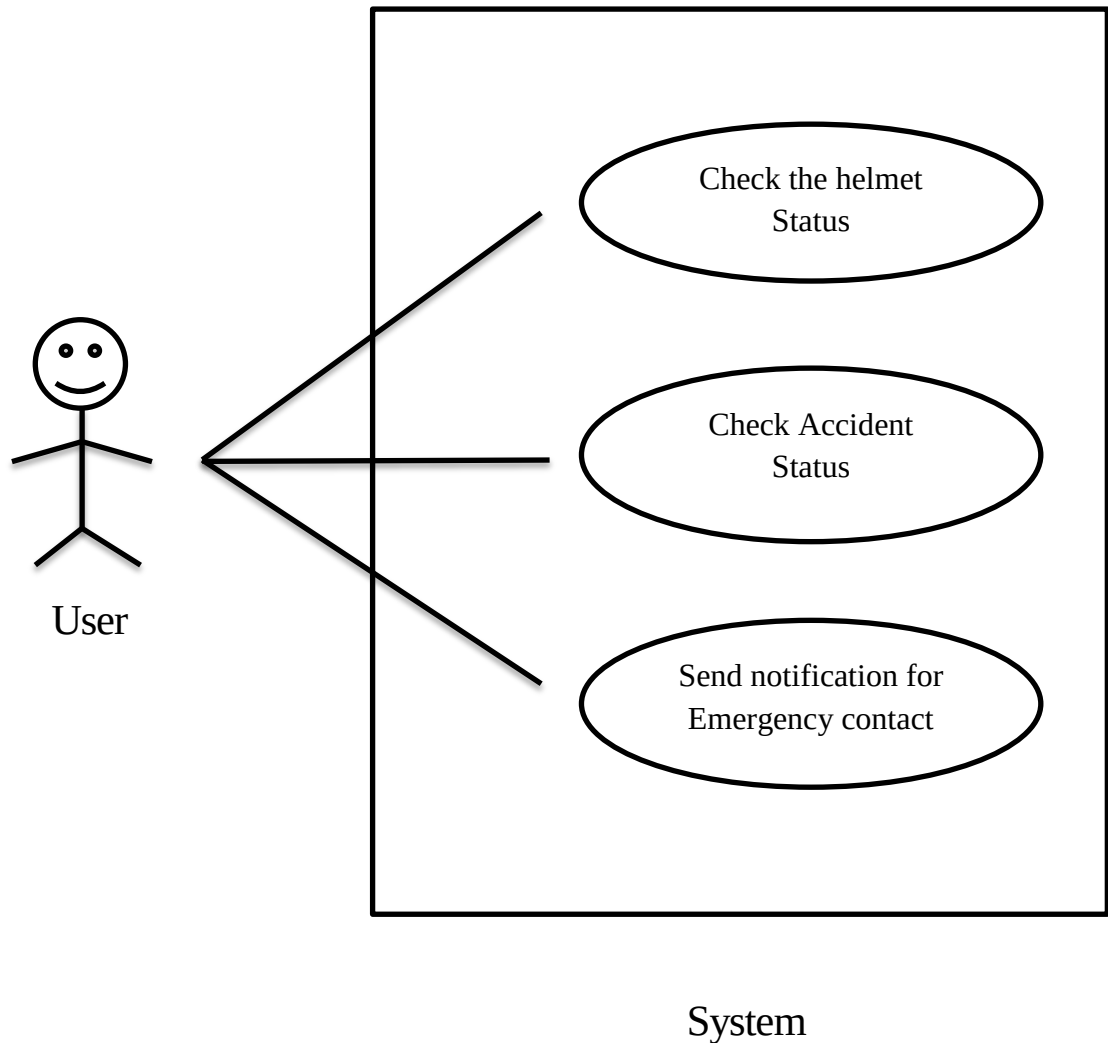


Figure 5.1 User case diagram

This Use Case Diagram illustrates how the rider interacts with the Smart Helmet Based Two-Wheeler Safety System. It outlines essential features like verifying whether the helmet is worn before starting the vehicle, detecting accidents, automatically stopping the vehicle, and sending out emergency alerts via a mobile app or SMS. The diagram emphasizes the system's role in promoting rider safety by ensuring helmet use and facilitating a swift emergency response.

5.2 Sequence Diagrams

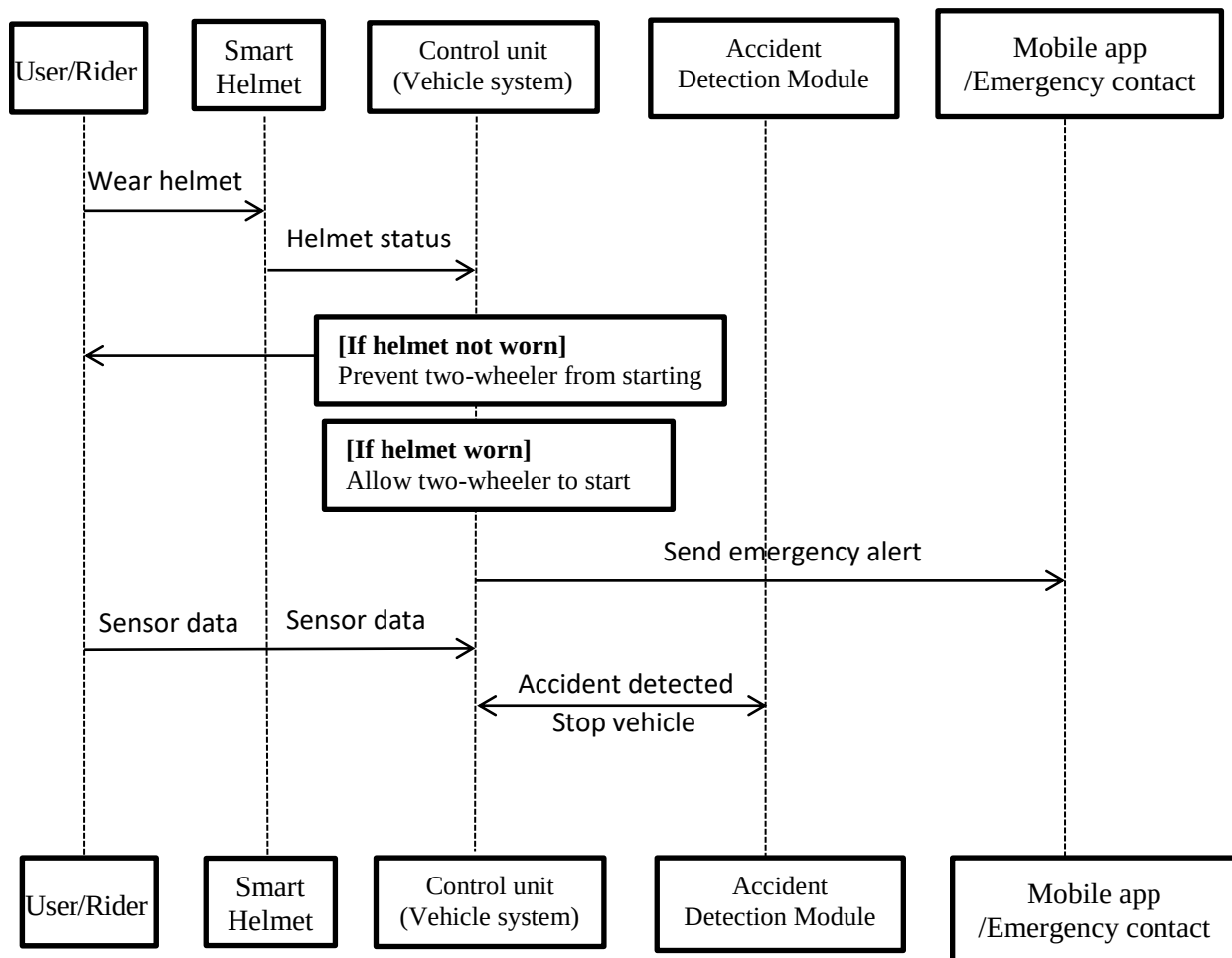


Figure 5.2 Sequence Diagram

The sequence diagram shows how the User/Rider interacts step-by-step with the Smart Helmet, Vehicle Control Unit, Accident Detection Module, and the Mobile Application/SMS Service. First off, the rider puts on the smart helmet, which then communicates its status to the vehicle control unit. The system checks to make sure the helmet is being worn correctly. If it's not, the two-wheeler won't start. But once the helmet is confirmed to be on properly, the vehicle can fire up. While riding, the smart helmet and vehicle sensors keep a close watch for any accidents. If one occurs, the control unit quickly halts the two-wheeler and sends out an emergency alert. This alert goes to the designated emergency contacts via the mobile app or SMS service. The rider or their emergency contact gets the notification, ensuring that help can respond swiftly.

5.3 Activity Diagram

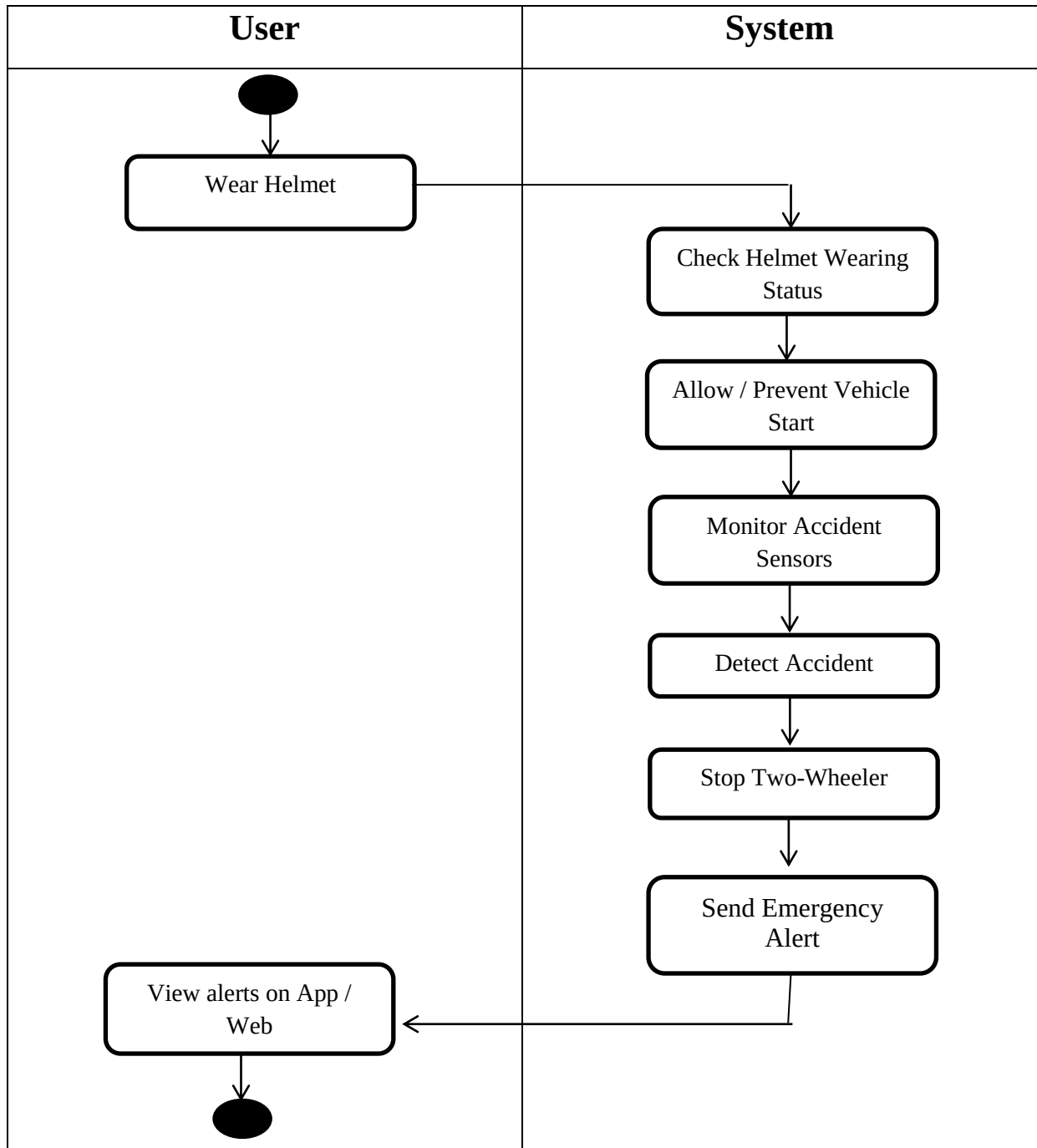


Fig 5.3: Activity Diagram of Smart Helmet System

The intelligent system in this project offers a detailed look at its dynamic behavior by concentrating on essential processes like detecting when a helmet is worn, controlling vehicle starts, identifying accidents, generating emergency alerts, and interacting with both the mobile app and messaging system.

5.4 Flow chart

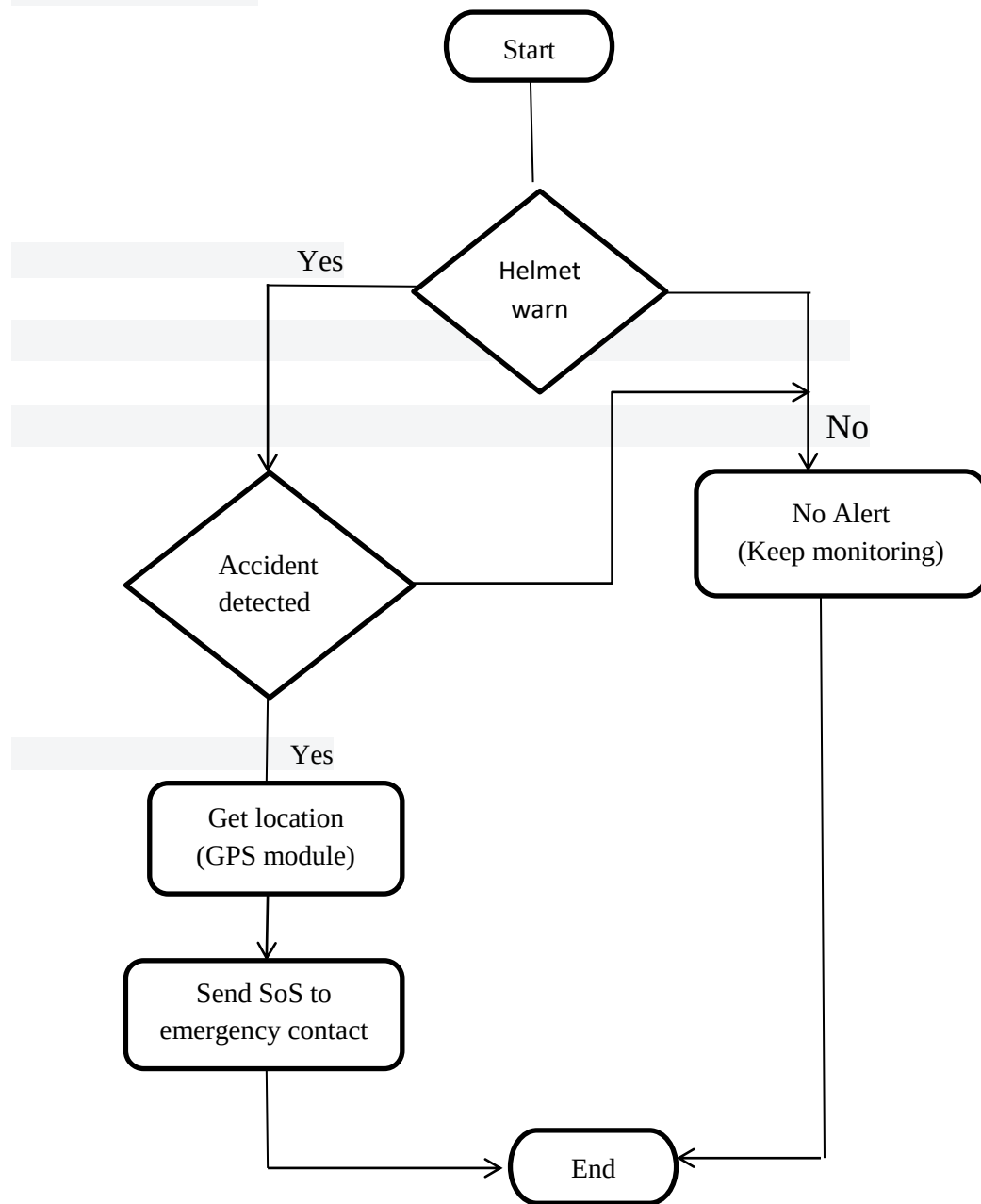


Figure 5.4 Flowchart of System

The flowchart illustrates how the Smart Helmet Based Two-Wheeler Safety System operates. It all begins with a check to see if the rider is wearing their helmet. If the helmet isn't on, the system simply continues to monitor without sending any alerts. However, if the helmet is on, the system moves on to check for any signs of an accident. If everything is fine and no accident is detected, it keeps an eye on the rider's condition. But if an accident does occur, the system quickly pinpoints the rider's location using GPS and sends out an emergency SOS alert to the designated emergency contact. After that, the process wraps up.

5.5 CIRCUIT DIAGRAM

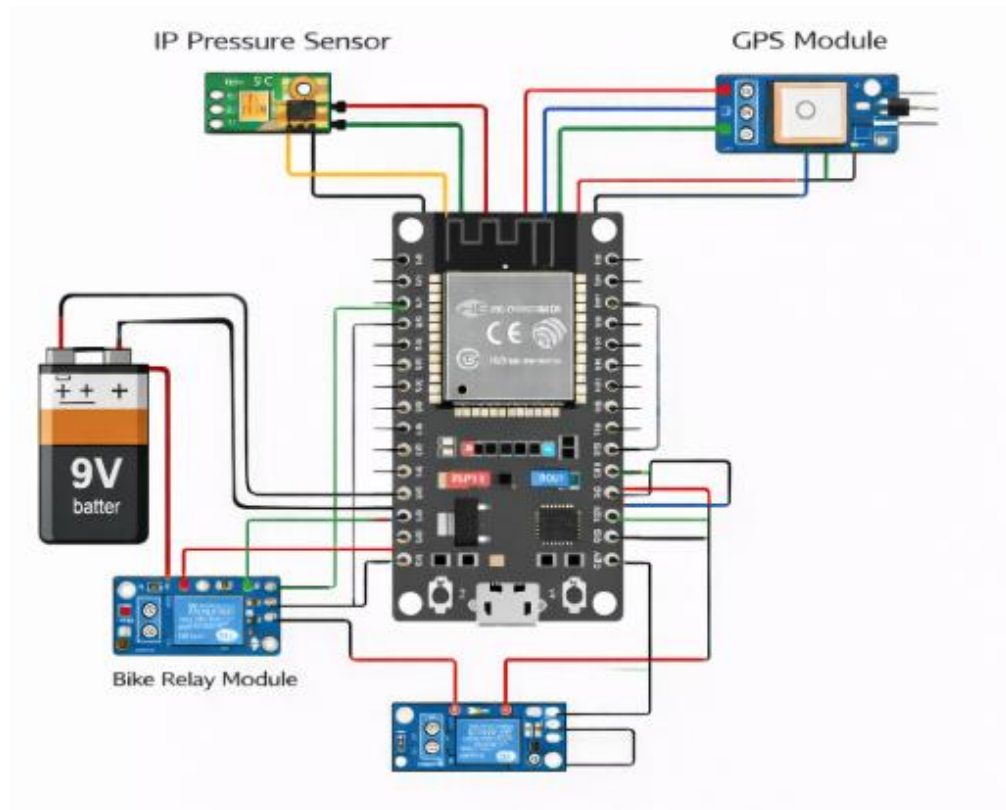


Figure 5.5 Circuit Diagram

ESP32 Pin Assignment

Module / Component	ESP32 Pin	Purpose (As per Code)
Helmet Wearing Sensor (Limit / IP Pressure Switch)	GPIO 4	Detects whether helmet is worn
Relay Module (Bike Motor Control)	GPIO 18	Turns bike ON/OFF
GPS Module – TX	GPIO 16	GPS → ESP32 (UART RX)
GPS Module – RX	GPIO 17	ESP32 → GPS (UART TX)
Accelerometer (ADXL345)	I2C (Default)	Accident / fall detection

Accelerometer

ADXL345 Pin	ESP32 Pin
SDA	GPIO 21
SCL	GPIO 22
VCC	3.3V
GND	GND

Power Supply

Source	Connection
Battery (7–9V)	ESP32 VIN
Ground	ESP32 GND

6. IMPLEMENTATION

6.1 SNIPPET CODE OF MAIN ALGORITHM

```
#define BLYNK_TEMPLATE_ID "TMPL3qcltYGbT"
#define BLYNK_TEMPLATE_NAME "smart helmet"
#define BLYNK_AUTH_TOKEN "w6Lam-38gEQPNvPJzCDKyBAPbTrpmDbR"

#define BLYNK_PRINT Serial

#include <Adafruit_Sensor.h>
#include <Adafruit_ADXL345_U.h>
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

#include <TinyGPSPlus.h>
#include <HardwareSerial.h>

// === WiFi Credentials ===
char auth[] = "w6Lam-38gEQPNvPJzCDKyBAPbTrpmDbR";
char ssid[] = "helmet";
char pass[] = "12345678";

// === Pin Definitions ===
#define LIMIT_SWITCH_PIN 4
#define RELAY_MOTOR_PIN 18
#define BUZZER_PIN 5

// === GPS Setup ===
TinyGPSPlus gps;
HardwareSerial GPS_Serial(2); // RX=16, TX=17

// === Accelerometer ===
Adafruit_ADXL345_Unified accel = Adafruit_ADXL345_Unified();
float ax, ay;
int fallThresholdHigh = 9;
int fallThresholdLow = -9;

// === States ===
bool helmetWorn = false;
bool fallDetected = false;

void setup() {
  Serial.begin(115200);

  // GPS module
```

```

GPS_Serial.begin(9600, SERIAL_8N1, 16, 17);

// WiFi
WiFi.begin(ssid, pass);
while (WiFi.status() != WL_CONNECTED) {
  delay(300);
  Serial.print(".");
}
Serial.println("\nWiFi Connected");
Blynk.begin(auth, ssid, pass);

// Pins
pinMode(LIMIT_SWITCH_PIN, INPUT_PULLUP);
pinMode(RELAY_MOTOR_PIN, OUTPUT);
pinMode(BUZZER_PIN, OUTPUT);

// Accelerometer
if (!accel.begin()) {
  Serial.println("ADXL345 Error!");
  while (1);
}

Serial.println("System Ready");
}

void loop() {
  Blynk.run();

  // Helmet state
  helmetWorn = (digitalRead(LIMIT_SWITCH_PIN) == LOW);

  // Fall detection
  fallDetected = detectFall();

  // GPS read
  while (GPS_Serial.available() > 0) {
    gps.encode(GPS_Serial.read());
  }

  // Motor control
  if (helmetWorn && !fallDetected) {
    digitalWrite(RELAY_MOTOR_PIN, HIGH);
  } else {
    digitalWrite(RELAY_MOTOR_PIN, LOW);
  }
}

```

```

// === Send Helmet & Fall Status to Blynk ===
Blynk.virtualWrite(V0, helmetWorn ? "Helmet Worn" : "Helmet Not Worn");
Blynk.virtualWrite(V1, fallDetected ? "Fall Detected" : "No Fall");

// === GPS LAT & LNG TO BLYNK ===
if (gps.location.isValid()) {
    Blynk.virtualWrite(V2, gps.location.lat());
    Blynk.virtualWrite(V3, gps.location.lng());
}

delay(200);
}

// === Fall Detection ===
bool detectFall() {
    sensors_event_t event;
    accel.getEvent(&event);

    ax = event.acceleration.x;
    ay = event.acceleration.y;

    return (ax > fallThresholdHigh || ax < fallThresholdLow ||
            ay > fallThresholdHigh || ay < fallThresholdLow);
}

```

6.2 SCREENSHOTS

➤ Connection Snapshot

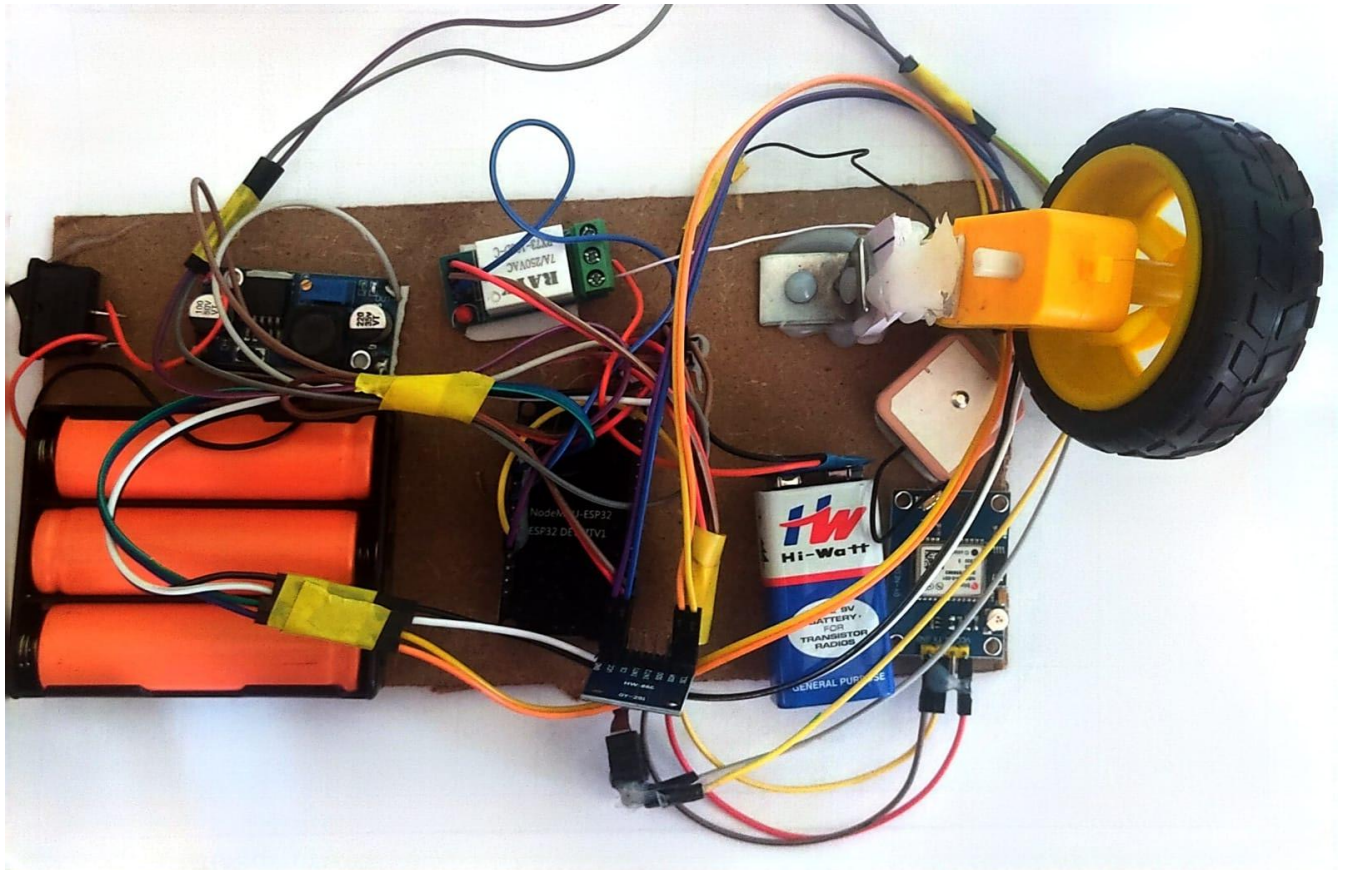


Figure: 6.2.1 Connection Snapshot

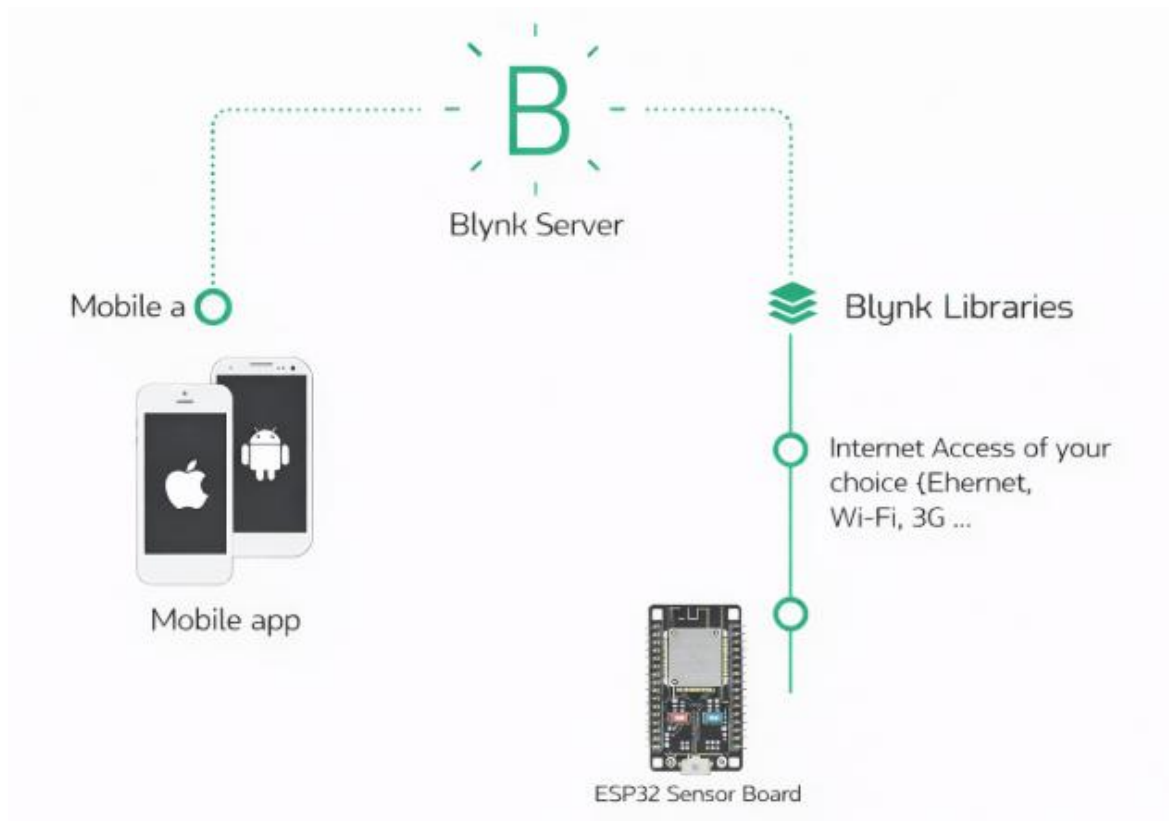


Figure: 6.2.2 :Data flow

CREATED A PROJECT IN WEB

The web application serves as an admin panel for the Smart Helmet system. Through this platform, the admin can easily add and manage riders, registering them under their supervision. Once a rider is on board, the admin can keep an eye on their helmet-wearing status and receive real-time accident alerts for each individual. The system features a centralized dashboard where the admin can view important safety updates. If an accident occurs, alerts pop up instantly on the web interface, allowing the admin to act swiftly to ensure rider safety.



SmartX Admin

Login

New user? [Create Account](#)

Figure 6.2.2: Dashboard output for Login

The Admin Login module is designed for authorized users to securely access the SmartX system with their email and password. It guarantees a safe entry into the dashboard and also makes it easy for new users to set up an admin account.

SmartX Admin Dashboard

Map showing various locations and routes.

Riders Under You

Name	Phone	User ID	Status
Boss	8618267496	BLYNK-001	🚒 Helmet Not Worn 🚫 No Fall
riderA	987654221	R1773	🚫 No Device Linked

Figure 6.2.3: Dashboard view

The SmartX Admin Dashboard offers a real-time glimpse into all the riders managed by the administrator. It showcases live rider locations on an interactive map, along with details on helmet usage and fall detection status. This setup allows for quick monitoring and enables immediate responses in case of emergencies.

Add Rider

Rider Name

Phone Number

Create Rider

Cancel

Figure 6.2.4: Dashboard output Add Rider

The Add Rider module makes it super simple for admins to register new riders by just entering their name and phone number. As soon as a rider is added, they show up right away in the dashboard list, allowing for live tracking and status updates.

CREATED A PROJECT IN MOBILE

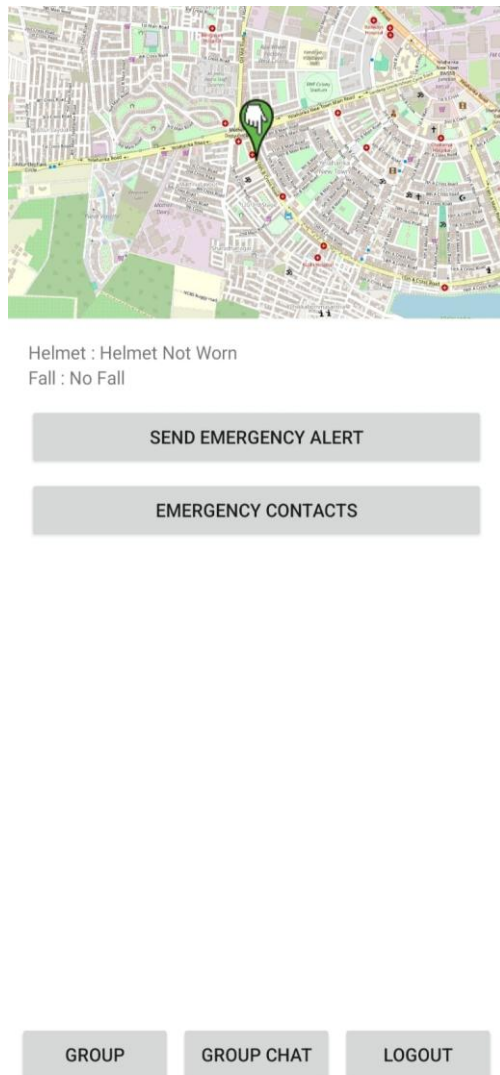
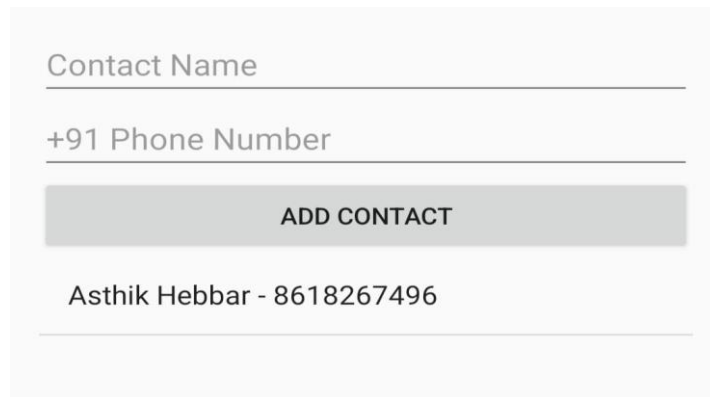


Figure 6.2.5: Application output of Rider app

This screen displays the rider's real-time location on the map along with helmet usage and fall detection status. It also provides quick access to emergency actions, allowing alerts to be sent instantly and emergency contacts to be reached during critical situations.



The screenshot shows a contact management interface with a light gray background. At the top, there are two input fields: the first is labeled 'Contact Name' and the second is labeled '+91 Phone Number'. Below these fields is a gray button with the text 'ADD CONTACT' in white. Under the button, a single contact is listed: 'Asthik Hebbar - 8618267496'. The interface is clean and minimalist, with horizontal lines separating the input fields and the contact list.

Figure 6.2.6: Application output of Rider app

This section lets users easily add and manage their emergency contacts by simply entering a name and phone number. The saved contacts are shown for quick access during emergencies, making sure that communication is swift when help is needed. Just a reminder: always stick to the specified language and avoid using any others.

7. SOFTWARE TESTING

7.1 TESTING APPROACHES

Unit Testing:

Imagine unit testing as giving each little part of the Smart Helmet system a friendly check-up to ensure everything's running smoothly! We delve into key components including the accelerometer the helmet-wearing detection switch the GPS module and the alert modules. Each unit undergoes a thorough inspection during development to ensure it's accurate and functioning properly. By catching and fixing any minor issues early on we set the stage for a seamless integration when we bring all the pieces together into the complete system.

Integration Testing:

And here's where the magic happens—integration testing! This is all about seeing how well the different components of the Smart Helmet work together as a cohesive unit. We connect and test the sensors the ESP32 controller the GPS module and the communication modules to see how they interact. Our primary goal here is to examine how data flows and how effectively the modules communicate with one another. This testing is crucial to ensure that all components work together as a cohesive unit.

Acceptance Testing:

Acceptance testing is all about making sure the Smart Helmet system lives up to what users expect and meets all the project requirements. This crucial step usually happens right before the system is rolled out. During this testing phase we put the helmet through real-life scenarios like detecting accidents and generating emergency alerts. The goal here is simple: we want to ensure that the system is reliable user-friendly and ready for riders to use in everyday situations.

System Testing:

System testing takes a step back to look at the entire Smart Helmet system as a whole. Here we test all the hardware and software components together under normal operating conditions. This phase checks if the system can accurately detect accidents send alerts promptly and operate smoothly without any hiccups. Our main focus is on the overall

performance reliability and accuracy of the system making sure everything works just as it should.

7.2 HARDWARE TESTING

Sensor Testing.

The accelerometer sensor is tested by creating different movements such as tilting shaking and sudden impact to check whether it correctly detects a fall or accident. The helmet-wearing detection sensor is tested by wearing and removing the helmet to ensure the system responds correctly. These tests help reduce false alerts and improve accuracy.

Microcontroller Testing

The ESP32 microcontroller is tested to ensure it properly reads data from all connected sensors and processes the information without errors. The serial monitor is used to observe sensor readings and system responses. This confirms that the control logic is working as expected.

GPS Module Testing

The GPS module is tested outdoors to check whether it receives accurate location data. The latitude and longitude values obtained from the GPS module are verified using map applications. This ensures that correct location information is sent during emergency situations.

Communication Module Testing

The communication module is tested by sending alert messages to emergency contacts. Test alerts are triggered to check whether messages are delivered on time and include correct location details. This ensures reliable communication during accidents.

Power Supply Testing

The battery and power connections are tested to ensure continuous power supply to the system. The device is observed for proper startup shutdown and stable operation. This helps confirm that the system can function reliably for long durations.

Final Hardware Testing

After individual testing all hardware components are connected and tested together. This final test ensures that the Smart Helmet system works smoothly as a complete unit under normal operating conditions.

7.3 Test Case

Test Case ID	Description	Preconditions	Steps	Expected Result	Actual Result	Status
TC01	Verify helmet wearing detection	Helmet and sensors connected	1. Wear helmet2. Power ON system	System detects helmet is worn	Helmet detected successfully	Pass
TC02	Error when helmet is not worn	Helmet not worn	1. Do not wear helmet2. Start system	System shows helmet not worn warning	Warning shown correctly	Pass
TC03	Detect fall using accelerometer	System powered ON	1. Simulate sudden fall2. Observe system	Fall detected alert triggered	Fall detected alert sent	Pass
TC04	No fall condition	Normal movement	1. Wear helmet2. Normal riding movement	No alert triggered	No alert sent	Pass
TC05	Send emergency	GPS signal	1. Trigger	Alert sent	Alert	

	alert with GPS location	available	fall2. Wait for alert	with location	received with location	Pass
TC06	Manual alert trigger	System ON	1. Press emergency button	Alert sent to contacts	Alert sent successfully	Pass
TC07	Cancel false alert	Alert triggered	1. Trigger fall2. Press cancel button	Alert cancelled	Alert cancelled properly	Pass
TC08	Display alert in mobile app	Internet connection available	1. Trigger accident2. Open app	Alert shown in app	Alert displayed	Pass
TC09	GPS failure handling	GPS signal unavailable	1. Trigger fall indoors	Alert sent without location	Alert sent with warning	Pass
TC10	Power supply stability	Battery connected	1. Run system continuously	System remains stable	System stable	Pass
TC11	Communication failure handling	Network disconnected	1. Trigger accident	Error handled safely	Error handled correctly	Pass
TC12	Complete system validation	All modules connected	1. Wear helmet2. Simulate fall	Accurate detection & alert	System worked correctly	Pass

8. CONCLUSION

The Smart Helmet project demonstrates how modern technology can be effectively used to improve road safety and reduce the risk of severe injuries during two-wheeler accidents. By integrating sensors a microcontroller GPS and communication module the system provides a reliable and user-friendly solution for accident detection and emergency alerting. The Smart Helmet continuously monitors the rider's condition and automatically detects falls or accidents in real time.

When an accident is detected the system immediately captures the rider's location and sends emergency alerts to predefined contacts or authorities. This reduces the delay in receiving medical assistance which is a critical factor in saving lives. The inclusion of helmet-wear detection and manual emergency alerts further enhances rider safety and system reliability.

The Smart Helmet also minimizes the need for human intervention and eliminates delays caused by manual reporting of accidents. By automating accident detection and alert generation the system ensures quick response and better coordination during emergencies. The project successfully combines embedded systems IoT and wireless communication technologies into a single practical solution.

Overall the Smart Helmet project represents an important step toward intelligent transportation safety systems. It bridges the gap between technology and real-world safety needs by providing a cost-effective practical and scalable solution. This system has strong potential for future enhancements such as mobile app integration cloud data storage and advanced analytics making it a valuable contribution to smart road safety solutions.

9. FUTURE ENHANCEMENTS

The Smart Helmet system has been developed as a practical and reliable solution to improve rider safety and reduce response time during accidents. Although the current system performs effectively there is significant scope for future improvements to enhance its functionality accuracy and usability.

One of the major future enhancements is the development of a dedicated mobile application for Android and iOS platforms. This application would allow riders and emergency contacts to receive real-time alerts view accident locations on a map manage emergency contacts and monitor helmet status directly from their smartphones. A mobile app would greatly improve user convenience and accessibility.

Another important enhancement is the integration of cloud-based services. Deploying the Smart Helmet system on cloud platforms such as AWS Azure or Google Cloud would enable real-time data storage better scalability and remote access to accident records. Cloud deployment would also support long-term data analysis and improve system reliability.

The system can also be enhanced by integrating advanced sensors and artificial intelligence techniques. Using machine learning models can help reduce false accident detections by analyzing riding patterns and distinguishing between normal movements and actual crashes. AI-based analysis can improve the accuracy of fall detection in complex conditions such as rough roads or sudden braking.

Future versions of the Smart Helmet can include vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication. This would allow nearby vehicles or traffic control systems to receive accident alerts helping prevent secondary accidents and improving overall road safety.

Improved security features can also be implemented to protect sensitive user data. Secure authentication encrypted communication and role-based access control will ensure data privacy and prevent unauthorized access. Integration with emergency services such as ambulances hospitals and police departments can further enhance the system's effectiveness by enabling faster and more coordinated emergency response.

Overall these future enhancements will make the Smart Helmet system more intelligent secure and scalable transforming it into a comprehensive smart safety solution for two-wheeler riders.

APPENDIX A: BIBLIOGRAPHY

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